CHAPTER 5

MEASURES OF CENTRAL TENDENCY

CHAPTER ESSENCE AT A GLANCE:
Determine representative central values: Mathematical averages (Arithmetic Mean: direct, shortcut, step-deviation methods) and Positional averages (Median, Quartiles, and Mode with empirical formula).

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Central Tendency Representative Role:** A single central value around which data points cluster, providing a summary representation of the entire distribution.

• **Arithmetic Mean Mathematical Invariants:** Sum of deviations from mean is always zero; Sum of squared deviations is minimum; affected by extreme outlier values.

• **Positional Averages (Median and Quartiles):** Median splits sorted data into two equal halves; Quartiles split sorted data into four quarters (Q_1 lower quartile, Q_3 upper quartile), completely immune to extreme outliers.

• **Mode and Empirical Relationship:** Mode represents the value with highest frequency; in moderately asymmetric distributions, Mode = 3 Median - 2 Mean.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$\bar{x} = \frac{\sum f_i x_i}{N} = A + \frac{\sum f_i d_i'}{N} \times c \quad \left[d_i' = \frac{x_i - A}{c}\right]$$

Continuous Series Step-Deviation Arithmetic Mean

$$\text{Median } (M) = L_1 + \frac{\frac{N}{2} - cf}{f} \times h$$

Continuous Grouped Distribution Median Formula

$$\text{Mode } (Z) = L_1 + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

Modal Class Inspection Continuous Mode Formula

$$\text{Mode } (Z) = 3 \times \text{Median } (M) - 2 \times \text{Mean } (\bar{x})$$

Karl Pearson Empirical Central Tendency Relationship

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The sum of deviations of items from their arithmetic mean is ALWAYS zero: sum of ($x - x_{bar}$) = 0.

• **Alert:** Median is the most suitable average for open-ended classes and qualitative phenomena; mean cannot be computed for open-ended tables without assuming boundaries.

• **Alert:** If the modal class has equal adjacent frequencies or highest frequency at the boundary, use the empirical formula Mode = 3 Median - 2 Mean.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State two mathematical properties of the Arithmetic Mean. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Property 1:** The algebraic sum of deviations of all observations from their arithmetic mean is always equal to zero (Sum of $(x - mean) = 0$). **[1 Mark]**

• **Property 2:** The sum of the squares of deviations from the mean is minimum compared to deviations from any other value. **[1 Mark]**

2. Write the empirical relationship connecting Mean, Median, and Mode in a moderately skewed distribution. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Formula:** Mode = 3 * Median - 2 * Mean (or Mean - Mode = 3 * (Mean - Median)). **[1 Mark]**

• **Application:** Used to ascertain the Mode when the distribution is irregular, bimodal, or has ill-defined modal peak. **[1 Mark]**

3. The mean of 50 observations was found to be 40. Later, it was discovered that one value 48 was wrongly copied as 84. Find the corrected mean. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Incorrect Sum of Items:** Sum of X = N * Mean = 50 * 40 = 2,000. **[1 Mark]**

• **Corrected Sum:** Correct Sum = Incorrect Sum (2,000) - Wrong Value (84) + Correct Value (48) = 2,000 - 36 = 1,964. **[1 Mark]**

• **Corrected Mean:** Correct Mean = 1,964 divided by 50 = 39.28. **[1 Mark]**

4. Calculate the Median from the following frequency distribution: Marks: 0-10 (Freq: 4), 10-20 (Freq: 6), 20-30 (Freq: 10), 30-40 (Freq: 8), 40-50 (Freq: 2). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Cumulative Frequencies:** 4, 10, 20, 28, 30. Total N = 30. N divided by 2 = 15. **[1 Mark]**

• **Median Class:** Cumulative frequency containing 15 is 20; hence Median Class is 20-30 (L1 = 20, cf = 10, f = 10, h = 10). **[1 Mark]**

• **Median Calculation:** M = 20 + ((15 - 10) divided by 10) * 10 = 20 + 5 = 25 marks. **[1 Mark]**

5. Why is the Median preferred over the Arithmetic Mean when dealing with extreme income inequalities or open-ended distributions? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Positional Character:** Median depends on the middle position of sorted data, remaining completely unaffected by extreme outliers. **[1.5 Marks]**

• **Open-Ended Usability:** Median requires knowledge of only frequencies and the middle class, allowing computation even when open-ended classes ('Below 100') exist. **[1.5 Marks]**

6. In a moderately asymmetric distribution, the Mean is 28 and the Median is 30. Calculate the Mode using the empirical relationship. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Empirical Formula:** Mode = 3 * Median - 2 * Mean. **[1 Mark]**

• **Substitution:** Mode = (3 * 30) - (2 * 28) = 90 - 56. **[1 Mark]**

• **Result:** Mode = 34. **[1 Mark]**

7. Calculate the Arithmetic Mean by Step-Deviation Method from the following data: Class: 10-20 (Freq: 5), 20-30 (Freq: 8), 30-40 (Freq: 12), 40-50 (Freq: 10), 50-60 (Freq: 5). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Mid-Values (m):** 15, 25, 35, 45, 55. Let Assumed Mean A = 35. Common class width c = 10. **[1 Mark]**

• **Step Deviations (d' = (m - 35) divided by 10):** -2, -1, 0, +1, +2. **[1 Mark]**

• **Products (f * d'):** (5 * -2 = -10), (8 * -1 = -8), (12 * 0 = 0), (10 * +1 = +10), (5 * +2 = +10). Sum of f * d' = -18 + 20 = +2. Total N = 40. **[1 Mark]**

• **Mean Formula & Computation:** Mean = A + (Sum of f * d' divided by N) * c = 35 + (2 divided by 40) * 10 = 35 + 0.5 = 35.5. **[1 Mark]**